

SDN CONTROLLER EVALUATION

Proprietary and Extended Control Capabilities for Cisco Hardware

EXECUTIVE SUMMARY

For a network team requiring deep, programmatic control over network hardware—including fine-grained configuration of switch features and ASIC-level functionalities beyond standard OpenFlow—the framework that best fits this requirement for Cisco hardware specifically is **Cisco onePK (One Network Environment Platform Kit)**.

1. CISCO ONEPK ANALYSIS

Cisco onePK was developed specifically by Cisco to provide rich, programmatic access deep into Cisco's operating systems (IOS, IOS-XE, IOS-XR, and NX-OS).

- **Deep Hardware & Feature Integration:** Unlike standard OpenFlow, which abstracts network hardware into standardized flow tables, onePK provides set/get APIs for underlying OS and ASIC capabilities, including QoS queues, ACLs, routing tables, system telemetry, and hardware interfaces.
- **Extensible Software Development Kits (SDKs):** It offers native SDK bindings for C, Java, and Python, allowing developers to build applications that directly manipulate Cisco platform internals.
- **Granular Control Beyond OpenFlow:** It grants access to proprietary switch features and native hardware ASIC capabilities that are not exposed by open standard protocols.

2. ALTERNATIVE OPTIONS EVALUATION

- **Generic OpenFlow Controllers:** While OpenFlow provides standardized flow-table manipulation, it intentionally abstracts hardware away. It lacks mechanisms to configure proprietary Cisco ASIC features, hardware queuing, or Cisco-specific OS subsystem settings.
- **Plexxi:** Plexxi's controller architecture is focused on photonics-driven, affinity-based dynamic data center fabrics and custom optical hardware rather than providing fine-grained control over standard Cisco switches.

CONCLUSION

Cisco onePK is the targeted mechanism for achieving fine-grained, proprietary, and extended programmatic control over Cisco network hardware beyond standard OpenFlow capabilities.